Adaptive Wind Power Forecasting Using Concept Drift Detection

A Technical Summary & Engineering Evaluation Report | B.Tech Artificial Intelligence & Machine Learning

Institution: Panimalar Engineering College (Anna Univ.)

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Target Horizon: 1-Hour-Ahead Regressive Forecasting

Turbine Capacity: 2.0 MW (2000 kW Aggregated)

1. Problem Formulation and Motivation

Wind energy forecasting models traditionally deployed in grid control centers operate under a static stationarity assumption. However, environmental regimes shift seasonally, turbine components suffer blade erosion or derating, and nearby obstacles create directional wake deficits. In such events (concept drift), static model errors multiply rapidly due to the non-linear cubic wind-power relation $P \propto v^3$. Blind periodic retraining (e.g. weekly) wastes computation and impairs stability when stationary. This project introduces a lightweight, interpretable, drift-adaptive forecasting framework utilizing a Histogram Gradient Boosting Regressor (HistGBR), heteroscedastic residual standardisation, Page-Hinkley sequential change detection, and a 48-hour refractory cooldown controller.

2. Forecast Accuracy and Baseline Comparison (10 Random Seeds)

Model Strategy	MAE (kW)	RMSE (kW)	nRMSE (%)	Retrainings	Compute (s)
Static Baseline (Train Once)	134.7 ± 5.6	178.4 ± 5.6	8.92 ± 0.28%	0.0	0.00 s
Adaptive Framework (Proposed)	**33.7 ± 2.1**	**53.0 ± 3.7**	**2.65 ± 0.19%**	16.2 ± 6.6	1.98 ± 0.82 s
Weekly Retrain (Blind Schedule)	35.3 ± 1.3	62.7 ± 5.4	3.14 ± 0.27%	80.0	9.59 ± 0.53 s

Key Findings: Under concept drift (week 28), the adaptive framework reduced MAE from 134.7 kW to 33.7 kW (**75.0% error reduction**, paired t-test $p = 3.0 \times 10^{-13}$). It achieved an accuracy comparable to or better than blind weekly retraining (33.7 kW vs 35.3 kW) while requiring **4.9× fewer retrainings** (16.2 vs 80.0) and **4.9× less compute**. In the no-drift control scenario, the adaptive model raised zero false alarms, whereas weekly retraining degraded MAE by **+30.2%**.

3. Concept Drift Detection & Change Point Dynamics

• **Detection Delay:** Mean delay of 21.5 hours across 10 random seeds (median 15.5 h). The delay represents the physical arrival of the first strong-wind regime (> 8 m/s), during which turbine derating alters output.
• **False Alarms:** 0 false alarms before drift onset across all ten runs.
• **Input Distribution Monitors:** Kolmogorov-Smirnov (KS) test and Population Stability Index (PSI) applied to wind speed revealed that natural autocorrelation causes distribution shifts even without concept change (nominal KS p-values of 10^{-1} to 10^{-31}). Therefore, input tests must serve strictly as an advisory channel corroborated by degraded prediction accuracy.

4. Verification of Hypotheses & Engineering Recommendations

Hypothesis	Claim	Status	Evidence

H1	Drift-triggered retraining outperforms static model	**Supported**	MAE 134.7 → 33.7 kW (75.0% lower, p = 3e-13)
H2	In stationary streams, adaptive model raises 0 alarms	**Supported**	0 alarms in 10 seeds; static equals adaptive
H3	Matches weekly retraining with far fewer updates	**Supported**	33.7 vs 35.3 kW with 4.9x fewer retrainings
H4	Adaptation benefit scales with drift magnitude	**Supported**	Gain scales from 0% (mag 0.1) to 82.8% (mag 1.5)

Report generated autonomously by Adaptive Wind Power Forecasting System. Grounded in empirical results.